## 2.3 LEIBNITZ'S THEOREM

**Statement:** If $y = uv$, where $u$ and $v$ are two functions of $x$ whose $n^{th}$ derivatives are known, then

$$y_n = (uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + {}^nC_3 u_{n-3} v_3 + \ldots + {}^nC_n u v_n.$$

**Proof:** $y = uv$

Scanned with CamScanner

Differentiating w.r.t. $x$ successively,

$$y_1 = u_1 v + u v_1$$
$$= {}^1C_0 u_1 v + {}^1C_1 u v_1$$
$$y_2 = u_2 v + 2 u_1 v_1 + u v_2$$
$$= {}^2C_0 u_2 v + {}^2C_1 u_1 v_1 + {}^2C_2 u v_2$$

This shows that theorem is true for $n = 1$ and $n = 2$.

Let the theorem is true for $n = m$.

$$y_m = (uv)_m = {}^mC_0 u_m v + {}^mC_1 u_{m-1} v_1 + {}^mC_2 u_{m-2} v_2 + \ldots\ldots\ldots + {}^mC_m u v_m.$$

Differentiating $y_m$ w.r.t. $x$,

$$\frac{d}{dx} y_m = \frac{d}{dx}(uv)_m = {}^mC_0 (u_{m+1} v + u_m v_1) + {}^mC_1 (u_m v_1 + u_{m-1} v_2) + {}^mC_2 (u_{m-1} v_2 + u_{m-2} v_3)$$
$$+ \ldots\ldots\ldots + {}^mC_m (u_1 v_m + u v_{m+1})$$
$$= {}^mC_0 u_{m+1} v + \left({}^mC_0 + {}^mC_1\right) u_m v_1 + \left({}^mC_1 + {}^mC_2\right) u_{m-1} v_2 + \ldots\ldots\ldots + {}^mC_m u v_{m+1}$$
$$y_{m+1} = (uv)_{m+1} = {}^{m+1}C_0 u_{m+1} v + {}^{m+1}C_1 u_m v_1 + {}^{m+1}C_2 u_{m-1} v_2$$
$$+ \ldots + {}^{m+1}C_{m+1} u v_{m+1}. \qquad \left[\because {}^mC_{r-1} + {}^mC_r = {}^{m+1}C_r\right]$$

This shows that theorem is true for $n = m + 1$ also.

By mathematical induction, theorem is true for all integer values of $n$.

Hence, $y_n = (uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + {}^nC_3 u_{n-3} v_3 + \ldots + {}^nC_n u v_n$.

**Example 1: If $y = x^2 e^{2x}$, prove that $(y_n)_0 = 2^{n-2} n (n-1)$.**

**Solution:** $y = x^2 e^{2x}$

Differentiating $n$ times using Leibnitz's Theorem,

$$y_n = x^2 2^n e^{2x} + n \cdot 2x 2^{n-1} e^{2x} + \frac{n(n-1)}{2!} \cdot 2 \cdot 2^{n-2} e^{2x}$$

Putting $x = 0$,

$$(y_n)_0 = 2^{n-2} n (n-1)$$

**Example 2: If $u$ is a function of $x$, and $y = e^{ax} u$, prove that $D^n y = e^{ax} (D + a)^n u$, where $D = \frac{d}{dx}$.**

**Solution:** $y = e^{ax} u$

$$D^n (e^{ax} u) = (D^n e^{ax}) u + {}^nC_1 (D^{n-1} e^{ax}) (Du) + {}^nC_2 (D^{n-2} e^{ax}) (D^2 u)$$
$$+ {}^nC_3 (D^{n-3} e^{ax}) (D^3 u) + \ldots\ldots\ldots + e^{ax} (D^n u)$$

Scanned with CamScanner

**Example 3: Find the $n^{th}$ order derivative of**

**(i)** $y = x^2 e^{ax}$ **(ii)** $x^3 \cos x$ **(iii)** $x^2 e^x \cos x$ **(iv)** $x^2 \tan^{-1}$

**Solution:** (i) $y = x^2 e^{ax}$

Let $u = e^{ax}$, $v = x^2$

Differentiating $n$ times using Leibnitz's Theorem,

$$y_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \ldots + {}^nC_n u v_n$$

$$= a^n e^{ax} \cdot x^2 + n a^{n-1} e^{ax} \cdot 2x + \frac{n(n-1)}{2!} a^{n-2} e^{ax} \cdot 2$$

$$= e^{ax}[x^2 a^n + 2nxa^{n-1} + n(n-1)a^{n-2}]$$

(ii) $y = x^3 \cos x$

Let $u = \cos x$, $v = x^3$

Differentiating $n$ times using Leibnitz's Theorem,

$$y_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \ldots + {}^nC_n u v_n$$

$$= \cos\left(x + \frac{n\pi}{2}\right) x^3 + n \cos\left[x + \frac{(n-1)\pi}{2}\right] 3x^2$$

$$+ \frac{n(n-1)}{2!} \cos\left[x + \frac{(n-2)\pi}{2}\right] 6x + \frac{n(n-1)(n-2)}{3!} \cos\left[x + \frac{(n-3)\pi}{2}\right] 6$$

$$= x^3 \cos\left(x + \frac{n\pi}{2}\right) + 3nx^2 \cos\left[x + \frac{(n-1)\pi}{2}\right]$$

$$+ 3x\, n(n-1) \cos\left[x + \frac{(n-2)\pi}{2}\right] + n(n-1)(n-2) \cos\left[x + \frac{(n-3)\pi}{2}\right]$$

Scanned with CamScanner

**Example 6:** If $y = (x^2 - 1)^n$, prove that $(x^2 - 1)\, y_{n+2} + 2x\, y_{n+1} - n\,(n+1)\, y_n = 0$.

**Solution:** $y = (x^2 - 1)^n$

Differentiating w.r.t. $x$,

$$y_1 = n(x^2 - 1)^{n-1} \cdot 2x$$

$$(x^2 - 1)\, y_1 = n(x^2 - 1)^n 2x = 2nyx$$

Differentiating again w.r.t. $x$,

$$(x^2 - 1)\, y_2 + 2xy_1 = 2\,(ny_1 x + ny)$$

$$(x^2 - 1)\, y_2 + (2x - 2nx)\, y_1 = 2ny$$

Differentiating the above equation $n$ times using Leibnitz's Theorem,

$$(x^2 - 1)y_{n+2} + n \cdot 2xy_{n+1} + \frac{n(n-1)}{2!} \cdot 2y_n + (2x - 2nx)y_{n+1} + n.2(1-n)y_n = 2ny_n$$

$$(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0.$$

**Example 7:** If $y = \sin[\log(x^2 + 2x + 1)]$, prove that $(x+1)^2\, y_{n+2} + (2n+1)\,(x+1)\, y_{n+1} + (n^2 + 4)\, y_n = 0$.

**Solution:** $y = \sin[\log(x^2 + 2x + 1)] = \sin[\log(x+1)^2] = \sin[2\log(x+1)]$

Differentiating w.r.t. $x$,

$$y_1 = \cos[2\log(x+1)] \cdot \frac{2}{x+1}$$

$$(x+1)y_1 = 2\cos[2\log(x+1)]$$

Scanned with CamScanner

Differentiating again w.r.t. $x$,

$$(x+1)y_2 + y_1 = -2\sin[2\log(x+1)]\cdot\frac{2}{(x+1)}$$

$$(x+1)^2 y_2 + (x+1)\, y_1 = -4y$$

Differentiating the above equation $n$ times using Leibnitz's Theorem,

$$(x+1)^2 y_{n+2} + n\cdot 2(x+1)y_{n+1} + \frac{n(n-1)}{2!}\cdot 2y_n + (x+1)y_{n+1} + n\cdot y_n = -4y_n$$

$$(x+1)^2 y_{n+2} + (2n+1)(x+1)\, y_{n+1} + (n^2+4)\, y_n = 0.$$

Scanned with CamScanner

**Example 9:** If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$, prove that $x^2 y_{n+2} + (2n+1)xy_{n+1} + 2n^2 y_n = 0$.

**Solution:**

$$\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n = n\log\left(\frac{x}{n}\right)$$

$$\frac{y}{b} = \cos\left[n\log\left(\frac{x}{n}\right)\right]$$

$$y = b\cos\left[n\log\left(\frac{x}{n}\right)\right]$$

Differentiating w.r.t. $x$,

$$y_1 = -b\sin\left[n\log\left(\frac{x}{n}\right)\right]\cdot n\cdot\frac{1}{\frac{x}{n}}\cdot\frac{1}{n} = \frac{-bn}{x}\sin\left[n\log\left(\frac{x}{n}\right)\right]$$

$$xy_1 = -bn\sin\left[n\log\left(\frac{x}{n}\right)\right]$$

Scanned with CamScanner

Differentiating again w.r.t. $x$,

$$xy_2 + y_1 = -bn\cos\left[n\log\left(\frac{x}{n}\right)\right]n\cdot\frac{1}{\frac{x}{n}}\cdot\frac{1}{n}$$

$$= \frac{-bn^2}{x}\cos\left[n\log\left(\frac{x}{n}\right)\right]$$

$$x^2y_2 + xy_1 = -bn^2\cos\left[n\log\left(\frac{x}{n}\right)\right] = -n^2y$$

Differentiating the above equation $n$ times using Leibnitz's Theorem,

$$x^2y_{n+2} + n\cdot 2xy_{n+1} + \frac{n(n-1)}{2!}\cdot 2y_n + xy_{n+1} + ny_n = -n^2y_n$$

$$x^2y_{n+2} + (2n+1)xy_{n+1} + 2n^2y_n = 0$$

Scanned with CamScanner

**Example 24:** If $y = \tan^{-1} x$, prove that (i) $(x^2+1)\, y_{n+1} + 2nxy_n + n\,(n-1)\, y_{n-1} = 0$

(ii) $y_n(0) = 0$, if $n$ is even

$= (-1)^{\frac{n-1}{2}} (n-1)!$, if $n$ is odd.

**Solution:** (i) $y = \tan^{-1} x$

Differentiating w.r.t. $x$,

$$y_1 = \frac{1}{1+x^2}$$

$$(x^2+1)\, y_1 = 1 \qquad \ldots (1)$$

Differentiating again w.r.t. $x$,

$$(x^2+1)\, y_2 + 2x\, y_1 = 0 \qquad \ldots (2)$$

Scanned with CamScanner

Differentiating Eq. (1) $n$ times using Leibnitz's Theorem,

$$(x^2+1)y_{n+1} + n \cdot 2xy_n + \frac{n(n-1)}{2!} \cdot 2y_{n-1} = 0$$

$$(x^2+1)\,y_{n+1} + 2nxy_n + n\,(n-1)\,y_{n-1} = 0 \quad \ldots(3)$$

(ii) Putting $x = 0$ in Eqs (1), (2) and (3),

$$y_1(0) = 1, y_2(0) = 0$$

$$y_{n+1}(0) = -n(n-1)\,y_{n-1}(0) \quad \ldots(4)$$

Putting $n = 2, 3, 4, \ldots$ in Eq. (4),

$$y_3(0) = -2(2-1)\,y_1(0) = -2 = -(2!) = (-1)^{\frac{3-1}{2}}(2!)$$

$$y_4(0) = -3(3-1)\,y_2(0) = 0$$

$$y_5(0) = -4(4-1)\,y_3(0) = -4(3)(-2) = (-1)^2(4!) = (-1)^{\frac{5-1}{2}}(4!)$$

$$y_6(0) = -5(4)\,y_4(0) = 0$$

$$y_7(0) = -6(5)\,y_5(0) = -6(5)(-1)^2(4!) = (-1)^3(6!) = (-1)^{\frac{7-1}{2}}(6!)$$

In general,

$$y_n = 0, \text{ if } n \text{ is even}$$

$$= (-1)^{\frac{n-1}{2}}(n-1)!, \text{ if } n \text{ is odd.}$$

**Example 25: If $y = e^{m\cos^{-1}x}$, find $(y_n)(0)$.**

**Solution:** $y = e^{m\cos^{-1}x}$

Differentiating w.r.t. $x$,

$$y_1 = e^{m\cos^{-1}x}\left(\frac{-m}{\sqrt{1-x^2}}\right) \quad \ldots(1)$$

$$(1-x^2)\,y_1^2 = m^2 y^2$$

Differentiating again w.r.t. $x$

$$(1-x^2)\,2y_1 y_2 - 2xy_1^2 = 2m^2 yy_1$$

$$(1-x^2)\,y_2 - xy_1 = m^2 y \quad \ldots(2)$$

Differentiating the above equation $n$ times using Leibnitz's Theorem,

$$(1-x^2)\,y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2!}(-2)y_n - xy_{n+1} - ny_n = m^2 y_n$$

$$(1-x^2)\,y_{n+2} - (2n+1)\,xy_{n+1} - (n^2+m^2)\,y_n = 0 \quad \ldots(3)$$

Putting $x = 0$ in Eqs (1), (2) and (3),

$$y_1(0) = -me^{m\cos^{-1}0} = -me^{\frac{m\pi}{2}}, \quad y_2(0) = m^2\,y(0) = m^2 e^{\frac{m\pi}{2}}$$

$$y_{n+2}(0) = (n^2+m^2)\,y_n(0) \quad \ldots(4)$$

Scanned with CamScanner

Putting $n = 1, 2, 3, 4, \ldots$ in Eq. (4),

$$y_3(0) = (1^2 + m^2)\, y_1(0) = -me^{\frac{m\pi}{2}} (1^2 + m^2)$$

$$y_4(0) = (2^2 + m^2)\, y_2(0) = m^2 e^{\frac{m\pi}{2}} (2^2 + m^2)$$

$$y_5(0) = (3^2 + m^2)\, y_3(0) = -me^{\frac{m\pi}{2}} (1^2 + m^2)(3^2 + m^2)$$

$$y_6(0) = (4^2 + m^2)\, y_4(0) = m^2 e^{\frac{m\pi}{2}} (2^2 + m^2)(4^2 + m^2)$$

In general,

Even derivative, $$y_{2n}(0) = m^2 e^{\frac{m\pi}{2}} (2^2 + m^2)(4^2 + m^2)\ldots[(2n-2)^2 + m^2]$$

Odd derivative, $$y_{2n+1}(0) = -me^{\frac{m\pi}{2}} (1^2 + m^2)(3^2 + m^2)\ldots[(2n-1)^2 + m^2]$$

**96. If $x + y = 1$, prove that**

Scanned with CamScanner